Response to Referee 1

Mixing effects in radiative decays of heavy-strange axial-vector mesons within light-cone QCD sum rules

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Author-review note. Working draft, 19 October 2026. Statements about manuscript additions assume that the accompanying proposed changes have been inserted into Prism. The numerical input-sensitivity part of Comment 1 remains incomplete; its completion note below must be replaced before submission.

We thank the referee for the positive assessment of our work and for the constructive suggestions. We address the three comments below.

Comment 1

> *In describing the approach used to calculate the radiative widths of axial-vector meson states, the authors refer to their previous work on estimating the mixing angle of the $^1P_1$ and $^3P_1$ spin states (Phys. Rev. D 111, No. 1 (2025) 014005). In my view, it would be beneficial for the reader if the authors briefly outlined this approach and the main results in the present article, taking into account—however slight—the changes in the input parameter values (such as the quark condensate magnitude) used in these studies.*

Response. We thank the referee for this suggestion. We have added a short account of the mixing-angle determination after the two-point sum rules for the decay constants in Section II. In the two-state method of our earlier work, the angle is obtained by imposing a vanishing off-diagonal spin-one correlator in the rotated current basis. For the convention used in the present manuscript,

$$R(\theta) = \begin{pmatrix} \sin\theta & \cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix}, \qquad \left[R\widehat{\mathbf{\Pi}}R^T\right]_{12} = 0. \tag{1}$$

With a common Borel and continuum prescription for the matrix entering the angle extraction, this gives

$$\tan(2\theta) = -\frac{2\widehat{\Pi}_{AB}}{\widehat{\Pi}_{AA} - \widehat{\Pi}_{BB}}. \tag{2}$$

The added explanation also makes explicit that this relation does not require equal physical diagonal correlators. Their difference cancels when the two inverse-rotation relations are divided. The earlier results, $\theta_{D_s} = 26.6^\circ \pm 0.6^\circ$ and $\theta_{B_s} = 38.5^\circ \pm 0.1^\circ$, are the external angle inputs used in the present numerical analysis.

We also clarify a typographical omission in the earlier article: the normalization factor $1/(m_Q + m_s)$ was included in the calculations, but was inadvertently omitted from the printed tensor-current definition. Thus, the normalization used in the two calculations is the same.

The updated QCD inputs are a separate issue. In particular, the light-quark condensate changes from $-(1.65 \pm 0.15) \times 10^{-2}\,\mathrm{GeV}^3$ to $-(0.240 \pm 0.010)^3\,\mathrm{GeV}^3$ at 1 GeV, while the central value of $\langle\bar{s}s\rangle/\langle\bar{q}q\rangle$ remains 0.80 and its uncertainty changes from 0.20 to 0.10. The strange-quark mass values are quoted at different scales and must be compared after consistent scale evolution.

Author-review note. Complete this part using the original mixing-angle implementation: first reproduce the earlier-input baseline, then repeat with the updated condensates and other inputs at consistent scales, holding the OPE and auxiliary prescription fixed. Report the resulting angle shifts for both sectors and compare them with the quoted uncertainties. No statement that the effect is negligible has yet been established. The existing $\theta = 90^\circ$ decay-constant check does not supply this comparison.

Comment 2

> *It seems to me that it would be very interesting to obtain—within the framework of the approach proposed by the authors—the values of the radiative decay widths into the vector D_s^* and B_s^* meson states. Of course, the calculation of these widths might well warrant a separate paper. However, the authors should, in my view, note that measuring absolute widths is a far more challenging experimental task than measuring the ratio of radiative decay widths into two different final states. This holds particularly true for the P-wave states of B_s mesons. One would also expect the ratio of decay probabilities into $D_s\gamma$ and $D_s^*\gamma$ for the $D_{s1}(2536)$ to depend strongly on the mixing angle.*

Response. We thank the referee for this useful suggestion. We agree that ratios of radiative widths into pseudoscalar and vector final states are experimentally more accessible than absolute widths, particularly for the bottom-strange mesons, and may provide complementary information on the mixing angle.

We are currently preparing a separate study of radiative transitions into the vector states D_s^* and B_s^*. Following the referee's suggestion, that study will also present the ratios to the pseudoscalar channels and examine their mixing-angle dependence, including the $D_{s1}(2536)$ channel. We have added a brief statement at the end of the Conclusion explaining the experimental motivation and the scope of this forthcoming study.

Comment 3

> *The decay constants f_1 and f_2 for the physical states $D_{s1}(2460)$ and $D_{s1}(2536)$, obtained in the paper as a byproduct, are of independent interest; they allow for the calculation—within the factorization approximation—of the ratio of the branching fractions for the B-meson decays into $D_{s1}(2536)D^-$ and $D_{s1}(2460)D^-$. As evidenced by LHCb results (arXiv:2411.03399), this ratio is experimentally accessible, given that the branching fractions for the decays of both $D_{s1}(2460)$ and $D_{s1}(2536)$ into $D_s\pi^+\pi^-$ have already been measured. It would be interesting to present the dependence of the ratio f_2/f_1 on the mixing angle in the paper.*

Response. We thank the referee for pointing out this additional use of the decay constants and the experimental motivation. We have added a figure in Section III showing $f_1(\theta)$ and $f_2(\theta)$ separately for both the charm-strange and bottom-strange systems. This displays the quantities that enter our radiative-decay sum rules and makes their angle dependence explicit. We supply the requested ratio $f_2(\theta)/f_1(\theta)$ in Fig. R1 of this response. The ratio figure is kept in the response, while the manuscript contains the individual constants.

Both plots use the same fixed representative QCD and auxiliary inputs as the radiative-width angular scan. The two-point residues are recomputed at each angle, and the ratio is formed from the two constants at that same angle. For these illustrative input choices,

$$\left.\frac{f_2^{(D_s)}}{f_1^{(D_s)}}\right|_{26.6^\circ} \simeq 0.418, \qquad \left.\frac{f_2^{(B_s)}}{f_1^{(B_s)}}\right|_{38.5^\circ} \simeq 0.166. \tag{3}$$

These are fixed-input scan values, rather than separately propagated predictions for the ratios. The radiative-width scan already includes the angle dependence of the residues; we have clarified its description and caption to state this explicitly.

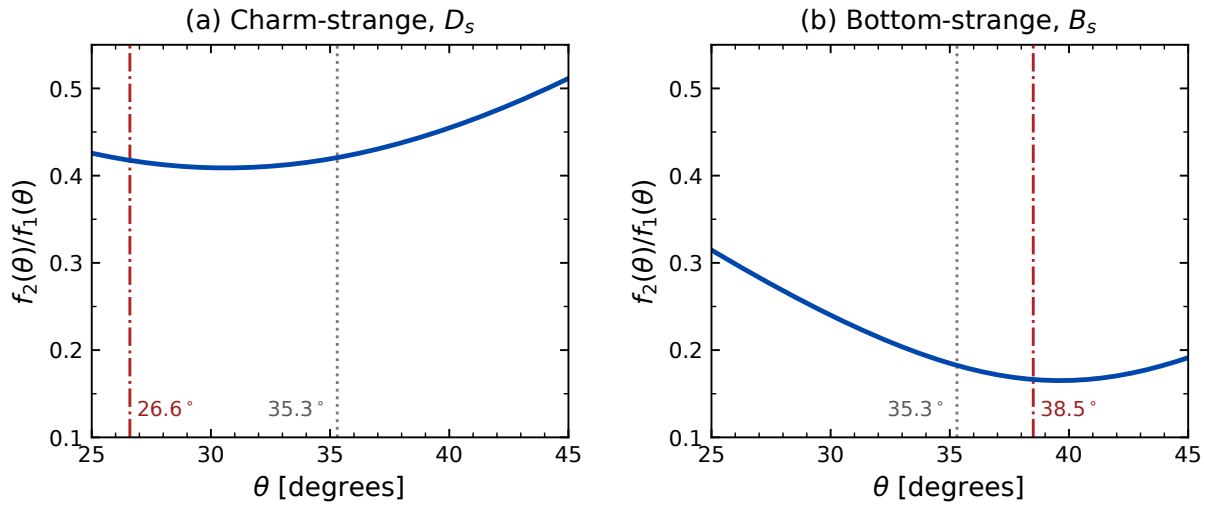


Figure R1: Ratio of the rotated-current decay constants, $f_2(\theta)/f_1(\theta)$, for the charm-strange (left) and bottom-strange (right) sectors. Both residues are evaluated at each angle with the same fixed representative inputs used for the individual-constant figure and the radiative-width angular scan. Red dash-dotted lines mark 26.6° and 38.5° in the corresponding panels; gray dotted lines mark the heavy-quark-limit reference 35.3°. No uncertainty bands are implied. This figure is provided for the referee response only.

For the suggested factorization application, a distinction between the currents is needed. Our f_i are defined by the rotated interpolating currents, $\langle 0|J_\mu^{(i)}|A_i\rangle = m_{A_i} f_i \eta_\mu$, whereas the nonleptonic weak amplitude involves the axial current $J_\mu^A = \bar{s}\gamma_\mu\gamma_5 c$. Within the adopted diagonal-coupling ansatz, the inverse current rotation gives

$$f_{A,1} = \sin\theta\, f_1, \qquad f_{A,2} = \cos\theta\, f_2, \qquad \frac{f_{A,2}}{f_{A,1}} = \cot\theta\, \frac{f_2}{f_1}. \tag{4}$$

The branching-fraction ratio therefore requires these weak-current constants together with the phase-space and $B \to D$ form-factor factors. This explains the usefulness of the referee's suggestion, while avoiding a direct identification of the branching-fraction ratio with $|f_2/f_1|^2$. A numerical analysis of the nonleptonic decays is beyond the scope of the present radiative-decay study.

References mentioned in this response

1. T. M. Aliev, S. Bilmis, and M. Savci, Phys. Rev. D **111**, 014005 (2025); arXiv:2408.08014.

2. LHCb Collaboration, "Study of $D_{s1}(2460)^+ \to D_s^+\pi^+\pi^-$ in $B \to \bar{D}^{(*)}D_s^+\pi^+\pi^-$ decays," Sci. Bull. **70**, 1432–1444 (2025), arXiv:2411.03399.